Fundamentals of AI and ML Project

Smart Study Planner

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INTRODUCTION

In today’s academic environment, students often have a hard time managing their study time effectively across different subjects. A good study plan is important to balance workload, improve performance, and lower stress.

This project, Smart Study Planner, aims to assist students in creating an optimized study schedule based on subject difficulty, past performance, and exam dates. The system uses basic machine learning techniques to predict how many hours a student should study each day for each subject.

OBJECTIVE

The main objectives of this project are:

- To create an intelligent study planner for students
- To use past study data to improve future predictions
- To generate a personalized daily study schedule
- To help students prepare effectively for exams

TECHNOLOGIES USED

- **Programming Language:** Python
- **Libraries Used:**
  - NumPy
  - scikit-learn (Linear Regression)
- **Data Storage:** JSON file
- **Concepts Used:**
  - Machine Learning
  - File Handling
  - Date & Time Handling

SYSTEM WORKING

The system works in the following steps:

Step 1: Add Subjects

The user inputs:

- Subject name
- Difficulty level (1–10)
- Past test score (%)
- Exam date

Step 2: Log Study Sessions

The user records how many hours they studied for a subject.
This data is stored and used for training the model.

Step 3: Train Machine Learning Model

If at least 3 study sessions are available:

- A Linear Regression model is trained
- Inputs (features): Difficulty, Past Score, Days until exam
- Output: Daily study hours

Step 4: Generate Study Plan

The system:

- Calculates days left until exam
- Predicts daily study hours
- Calculates total study time
- Displays a 3-day study schedule

MACHINE LEARNING APPROACH

The project uses **Linear Regression** to predict study time.

Input Features:

- Difficulty of subject
- Past performance score
- Days remaining until exam

Output:

- Recommended daily study hours

If insufficient data is available, a fallback formula is used based on:

- Difficulty
- Score
- Urgency (days left)

DATA STORAGE

All data is stored in a JSON file:

Subjects:

- Name
- Difficulty
- Score
- Exam date

Study Sessions:

- Subject name
- Difficulty
- Score
- Days until exam
- Hours studied

This allows the system to learn and improve over time.

FEATURES OF THE SYSTEM

- Add and remove subjects
- Log daily study sessions
- Intelligent study schedule generation
- Machine learning-based predictions
- Automatic calculation of study hours
- Simple command-line interface

ADVANTAGES

- Easy to use
- Personalized study planning
- Improves time management
- Learns from user behavior
- Helps reduce exam stress

LIMITATIONS

- Requires manual data input
- Accuracy depends on quality of logged sessions
- Basic machine learning model (can be improved)
- No graphical interface

FUTURE IMPROVEMENTS

- Add graphical user interface (GUI)
- Include timetable with time slots
- Add reminders and notifications
- Improve ML model accuracy
- Add performance tracking graphs

CONCLUSION

The Smart Study Planner is a useful tool for students to manage their study time effectively. By combining basic machine learning with user input, it provides personalized recommendations that improve with usage.

This project demonstrates how simple AI techniques can be applied to solve real-life problems like time management and exam preparation.

REFERENCES

- Python Documentation
- scikit-learn Documentation
- NumPy Documentation
- Basic Machine Learning Concepts